HDB Resale Price Prediction

Dhathathreyan Natarajan

Declaration: I used Generative AI tools (specifically Qwen) to refine texts, debug R code, and verify statistical reasoning. I am responsible for the content and quality of the submitted work, along with the code.

Part 1: Introduction

The objective of this report is to develop a statistically adequate linear regression model for predicting HDB resale prices in Singapore. The analysis uses 2,456 resale transactions from July 2020 (data.gov.sg), with 11 variables covering flat attributes (town, flat type, floor area, storey range, lease details) and resale price. Part I presents exploratory data analysis (EDA) to assess response distribution, predictor relationships, and variable types. Part II details the modelling pipeline: initial specification, Box-Cox transformation, model adequacy checking via residual diagnostics, stepwise variable selection using AIC, and final interpretation. The goal is a parsimonious, well-fitting model that satisfies standard regression assumptions and balances predictive accuracy with interpretability.

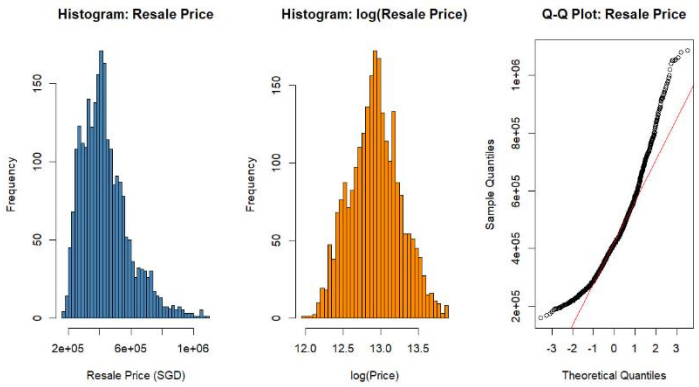
Part 2: Exploratory Data Analysis

The response variable is resale_price (SGD). Table 1 summarises its distribution. The raw price exhibits a mean of SGD 436,487 and median of SGD 413,000, with positive skewness of 1.04, indicating a right-skewed distribution with a long upper tail possibly driven by high-value transactions in prime locations. The normal Q-Q plot (Figure 1, right panel) further confirms this departure from normality. Observed quantiles deviate substantially from the theoretical reference line, particularly in the upper tail where points curve upward, indicating heavier right tails than expected under normality. The log-transformed price reduces skewness to approximately 0.12, closely approximating symmetry, suggesting a log transformation of the response might be appropriate. This is formally confirmed by Box-Cox analysis later in the report.

Table 1: Summary Statistics

Statistic	Value
Mean	SGD 436,487
Median	SGD 413,000
Std. Dev.	SGD 152,382
Min	SGD 160,000
Max	SGD 1,085,000
Skewness	1.04

Figure 1: Distribution and normal Q-Q Plot



Before model fitting, several derived variables were constructed from the raw data:

Continuous Variable Transformations:

remaining_lease_dec: The variable *remaining_lease* was parsed into decimal years.

storey_mid: The categorical *storey_range* was replaced by its numeric midpoint to preserve ordinal information while reducing dimensionality.

flat_age: Computed as 2020 − *lease_commence_date*, the age of the flat at the time of sale.

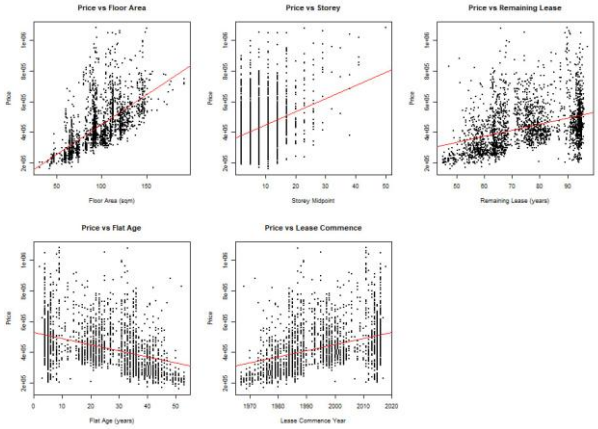
Sparse category handling: The single "Multi-generation" observation was merged with "Executive" to prevent a near-singular dummy variable. From a domain perspective, Multi-Generation flats are architecturally derived from the Executive layout, both being premium large units. This decision aligns with the parsimony principle, to maintain model stability and retain explanatory power. The variable *flat_model* (architectural design iteration) was excluded in favour of *flat_type* because the latter captures the economically meaningful size/prestige gradient with fewer parameters, avoids sparsity from rare model categories, and aligns with the analysis objective of identifying broad, actionable price drivers.

Figure 2 presents scatter plots of resale price against each continuous predictor. *floor_area_sqm* (31–192 sqm) shows the strongest positive linear association with price. *storey_mid* (2–50) and *remaining_lease_dec* (45.5–97.1 years) show moderate positive relationships. All three exhibit broadly linear trends with no obvious curvature, supporting the use of linear terms. The plots for

flat_age and *lease_commence_date* are mirror images of *remaining_lease_dec*, visually confirming their perfect collinearity.

Figure 3 shows boxplots of resale price by town and flat type. Substantial price variation exists across the 26 towns. This validates retaining *town* at its original granularity; grouping into broad geographic regions would mask economically meaningful within-region differences. *flat_type* shows a clear monotonic price increase from 1 Room to Executive, consistent with size being a primary price driver.

Figure 2: Scatter plot of response against continuous predictors / Figure 3: Boxplot of response against town/type



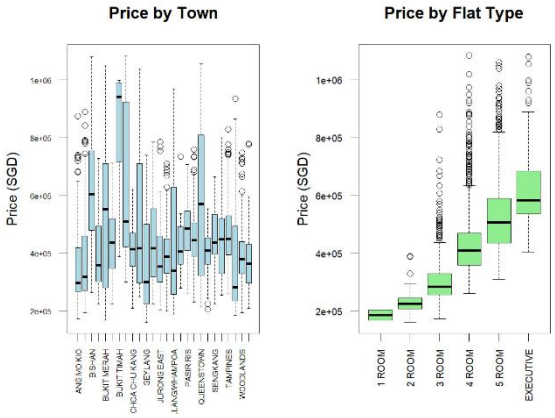
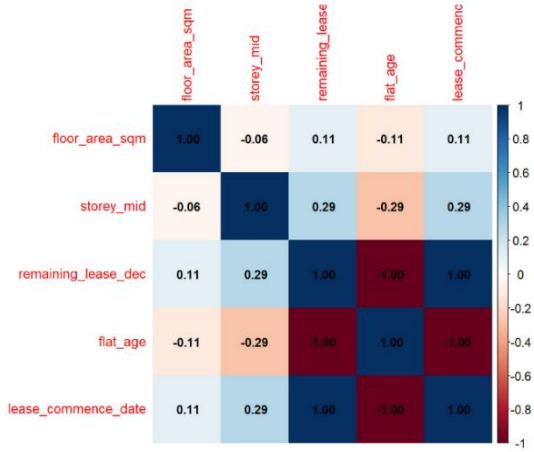


Figure 6: Summary Table (Initial Model)



Perfect multicollinearity was detected among remaining_lease_dec, flat_age, and lease_commence_date (|r| = 0.9993; Figure 4), confirming the exclusion of the latter two variables.

Among the three retained continuous predictors (floor_area_sqm, storey_mid, remaining_lease_dec), pairwise correlations are low (|r| ≤ 0.29). The condition number computed from their correlation matrix is **κ = 1.94** (eigenvalues: 1.29, 1.04, 0.67), well below the threshold of 100, confirming no multicollinearity concern among retained predictors.

Variance Inflation Factors (VIFs) are assessed post-model-fitting in Part 4 to evaluate residual collinearity involving categorical predictors.

Part 3: Model Building

The initial model M_0 regresses *resale_price* directly on the predictors without any transformation, serving as the baseline for comparison. M_0 is specified as:

$M_0 : resale_price = \beta_0 + \beta_1\, floor_area_sqm + \beta_2\, storey_mid + \beta_3\, remaining_lease_dec + \beta_4\, town + \beta_5\, flat_type + \varepsilon$. This baseline specification serves to;

(i) Establish preliminary fit statistics,
(ii) Identify potential model inadequacies requiring transformation
(iii) Provide a reference point for comparing alternative specifications

Figure 5: Anova Table (Initial Model)

Analysis of Variance Table

Response: resale_price

	Df	Sum Sq	Mean Sq	F value	Pr(>F)	
floor_area_sqm	1	2.3189e+13	2.3189e+13	6259.5049	< 2.2e-16	***
storey_mid	1	7.2345e+12	7.2345e+12	1952.8718	< 2.2e-16	***
remaining_lease_dec	1	2.5955e+12	2.5955e+12	700.6163	< 2.2e-16	***
town	25	1.4861e+13	5.9443e+11	160.4590	< 2.2e-16	***
flat_type	5	1.5447e+11	3.0895e+10	8.3397	7.903e-08	***
Residuals	2422	8.9724e+12	3.7045e+09			

The ANOVA table (Figure 5) confirms that all five predictor groups are jointly highly significant ($p < 2.2e\text{-}16$). All three continuous predictors are individually significant ($p < 2e\text{-}16$). Most town dummies are significant; Bedok, Geylang, and Serangoon are individually insignificant but the group F-test confirms joint significance ($F = 160.46$, $p < 2.2e\text{-}16$).

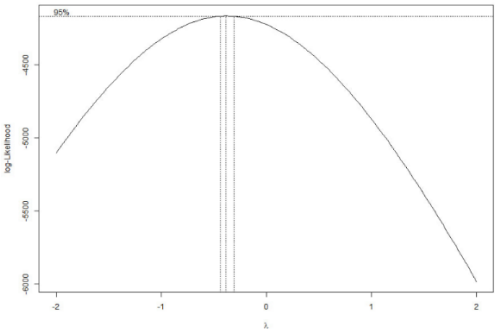
The *flat_type* dummies for 2 Room through 5 Room are individually insignificant, while Executive is significant (p = 0.009); the group jointly is significant (F = 8.34, p < 0.001), indicating that flat type as a group contributes meaningfully to explaining price. M_0 achieves Adj-R^2 = 0.8405 with residual standard error of SGD 60,860. The residuals range from −SGD183,011 to +SGD434,036, suggesting heteroscedasticity and non-normality (Shapiro-Wilk W = 0.9607, p < 0.001). These issues motivate a response transformation, assessed formally via Box-Cox.

Figure 6: Coefficients Table (M_0)

```
Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)       -372358.5    45125.9  -8.252 2.54e-16 ***
floor_area_sqm       3732.4      187.2  19.936  < 2e-16 ***
storey_mid           4451.5      239.4  18.591  < 2e-16 ***
remaining_lease_dec  5527.3      123.6  44.725  < 2e-16 ***
townBEDOK           -9792.0     7969.6  -1.229  0.21932
townBISHAN         121269.4    13164.8   9.212  < 2e-16 ***
townBUKIT BATOK    -91309.8     9301.7  -9.816  < 2e-16 ***
townBUKIT MERAH    104693.0     9677.6  10.818  < 2e-16 ***
townBUKIT PANJANG -158364.7     8612.6 -18.388  < 2e-16 ***
townBUKIT TIMAH    228119.6    25832.7   8.831  < 2e-16 ***
townCENTRAL AREA   189734.1    15402.6  12.318  < 2e-16 ***
townCHOA CHU KANG -189285.9     8857.4 -21.370  < 2e-16 ***
townCLEMENTI        41987.5     9494.3   4.422 1.02e-05 ***
townGEYLANG         19466.5    10003.9   1.946  0.05178 .
townHOUGANG        -72353.8     8376.0  -8.638  < 2e-16 ***
townJURONG EAST    -89849.4    10476.7  -8.576  < 2e-16 ***
townJURONG WEST   -130277.7     8192.7 -15.902  < 2e-16 ***
townKALLANG/WHAMPOA  69578.1    11366.5   6.121 1.08e-09 ***
townMARINE PARADE   123494.4    18708.4   6.601 5.00e-11 ***
townPASIR RIS      -106048.7     9155.6 -11.583  < 2e-16 ***
townPUNGGOL        -126681.5     8247.3 -15.360  < 2e-16 ***
townQUEENSTOWN      122322.0    10007.4  12.223  < 2e-16 ***
townSEMBAWANG      -208075.8    10364.2 -20.076  < 2e-16 ***
townSENGKANG       -158805.1     8224.6 -19.309  < 2e-16 ***
townSERANGOON       -10654.3    12205.8  -0.873  0.38281
townTAMPINES        -47173.2     7975.6  -5.915 3.79e-09 ***
townTOA PAYOH        24603.0     9390.1   2.620  0.00885 **
townWOODLANDS      -173397.9     8137.2 -21.309  < 2e-16 ***
townYISHUN         -130652.2     7992.8 -16.346  < 2e-16 ***
flat_type2 ROOM     18672.1    44434.7   0.420  0.67437
flat_type3 ROOM     51674.0    44273.5   1.167  0.24326
flat_type4 ROOM     67368.2    45414.2   1.483  0.13809
flat_type5 ROOM     86304.9    46753.8   1.846  0.06502 .
flat_typeEXECUTIVE 126910.3    48758.2   2.603  0.00930 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 60860 on 2422 degrees of freedom
Multiple R-squared:  0.8426,    Adjusted R-squared:  0.8405
F-statistic: 392.9 on 33 and 2422 DF,  p-value: < 2.2e-16
```

Figure 7: Box Cox Plot



A Box-Cox analysis was applied to M_0 where the procedure evaluates the log-likelihood over a grid of power transformation values λ. The optimal λ = −0.38, and the 95% confidence interval spans approximately −0.5 to −0.2, excluding both 0 and 1. This rules out no transformation and suggests a moderately strong negative power transformation. Two candidate transformations are evaluated: λ = 0 (log transformation) and λ = −0.5 (reciprocal square root). Two transformed models are fitted alongside and assessed against M_0 for comparison. The log-transformed model M_0_log uses log(*resale_price*) as the response, motivated by the right-skewed distribution observed in EDA and the interpretability of log-scale coefficients as percentage price changes. The reciprocal square root model M_0_bc uses 1/√(*resale_price*) as the response, corresponding to λ = −0.5 and representing the closest standard transformation to the Box-Cox optimal.

M_0_log: $\log(resale_price) = \beta_0 + \beta_1\, floor_area_sqm + \beta_2\, storey_mid + \beta_3\, remaining_lease_dec + \beta_4\, town + \beta_5\, flat_type + \varepsilon$

M_0_bc: $1/\sqrt{(resale_price)} = \beta_0 + \beta_1\, floor_area_sqm + \beta_2\, storey_mid + \beta_3\, remaining_lease_dec + \beta_4\, town + \beta_5\, flat_type + \varepsilon$

Table 2 summarises the fit statistics for all three models. M_0_bc achieves the highest Adj-R^2 (0.8922) and lowest AIC (−38,949), while M_0_log achieves Adj-R^2 = 0.8858 and AIC = −3,686. Both substantially outperform the untransformed M_0.

Table 2: Fit statistics for all 3 models

Model	**Response**	**Adj-R^2**	**AIC**	**Shapiro-Wilk W**
M_0	resale_price	0.8405	61,118	0.9607
M_0_log	log(resale_price)	0.8858	−3,686	0.9764
M_0_bc	1/√(resale_price)	0.8922	−38,949	0.9788

Despite M_0_bc having marginally better fit statistics, M_0_log is selected as the basis for further modelling. The reciprocal square root transformation does not permit meaningful economic interpretation of coefficients where a unit change in a predictor produces a change in 1/√(price), which cannot be translated into a practical price effect without back-transformation that is not intuitive to stakeholders. In contrast, the log model allows each coefficient to be interpreted directly as a percentage change in resale price, which is practically meaningful. The improvement in Shapiro-Wilk W from M_0 (0.9607) to M_0_log (0.9764) confirms that the log transformation also better satisfies the normality assumption. Model adequacy is assessed formally in the next section.

Residual diagnostics confirm that the log transformation resolves the inadequacies of the untransformed model. Compared to M_0, M_0_log exhibits markedly improved normality (Shapiro-

Wilk W increases from 0.9607 to 0.9764), eliminates the funnel pattern in residuals vs. fitted values, and removes systematic curvature in residuals vs. predictors. The reciprocal square root specification (M_0_bc) shows similar diagnostic improvements but offers no meaningful gain in fit (ΔAdj-R^2 = 0.0064) while sacrificing interpretability. Full residual diagnostics for the final model MN are presented in Part 4.

Part 4: Final Model Specification, Diagnostics, and Interpretation

4.1 Final Model Specification

Following the transformation selection in Part 3, the log-transformed response log(resale_price) is retained for final modelling. The initial specification includes all predictors identified in the exploratory phase:

$$\log(\text{resale_price}) = \beta_0 + \beta_1\text{floor_area_sqm} + \beta_2\text{storey_mid} + \beta_3\text{remaining_lease_dec} + \beta_4\text{town} + \beta_5\text{flat_type} + \varepsilon$$

4.2 Stepwise Variable Selection

To assess whether all predictors contribute meaningfully to model fit, stepwise selection via Akaike Information Criterion (AIC) was applied with bidirectional elimination:

Result: No variables were removed. All predictors (*floor_area_sqm, storey_mid, remaining_lease_dec, town,* and *flat_type*) are retained in the final model (denoted MN), indicating that each contributes to explaining price variation after accounting for all other variables.

The model explains 88.6% of the variance in log resale prices, representing a substantial improvement over the untransformed baseline (ΔAdj-R^2 = 0.0453). The overall model is highly significant (F = 578 on 33 and 2422 DF, p < 2.2e-16), confirming that the predictors jointly explain substantial variation in log resale prices as seen in Figure 8.

Figure 8: Coefficient (M_N)

```
Coefficients:
                          Estimate Std. Error t value Pr(>|t|)
(Intercept)             10.9219514  0.0840860 129.890  < 2e-16 ***
floor_area_sqm           0.0088384  0.0003489  25.335  < 2e-16 ***
storey_mid               0.0082703  0.0004462  18.537  < 2e-16 ***
remaining_lease_dec      0.0122890  0.0002303  53.365  < 2e-16 ***
townBEDOK               -0.0166716  0.0148503  -1.123 0.261700
townBISHAN               0.2145034  0.0245308   8.744  < 2e-16 ***
townBUKIT BATOK         -0.1793297  0.0173324 -10.346  < 2e-16 ***
townBUKIT MERAH          0.1982473  0.0180328  10.994  < 2e-16 ***
townBUKIT PANJANG       -0.2983826  0.0160483 -18.593  < 2e-16 ***
townBUKIT TIMAH          0.3202700  0.0481356   6.653 3.53e-11 ***
townCENTRAL AREA         0.3603472  0.0287006  12.555  < 2e-16 ***
townCHOA CHU KANG       -0.3710261  0.0165046 -22.480  < 2e-16 ***
townCLEMENTI             0.0927561  0.0176913   5.243 1.72e-07 ***
townGEYLANG              0.0102417  0.0186409   0.549 0.582768
townHOUGANG             -0.1375393  0.0156076  -8.812  < 2e-16 ***
townJURONG EAST         -0.1741945  0.0195219  -8.923  < 2e-16 ***
townJURONG WEST         -0.2549861  0.0152660 -16.703  < 2e-16 ***
townKALLANG/WHAMPOA      0.1150645  0.0211800   5.433 6.10e-08 ***
townMARINE PARADE        0.3339706  0.0348606   9.580  < 2e-16 ***
townPASIR RIS           -0.1854421  0.0170603 -10.870  < 2e-16 ***
townPUNGGOL             -0.2298919  0.0153678 -14.959  < 2e-16 ***
townQUEENSTOWN           0.2241733  0.0186473  12.022  < 2e-16 ***
townSEMBAWANG           -0.3979609  0.0193122 -20.607  < 2e-16 ***
townSENGKANG            -0.2943437  0.0153254 -19.206  < 2e-16 ***
townSERANGOON            0.0016817  0.0227439   0.074 0.941063
townTAMPINES            -0.0663124  0.0148614  -4.462 8.49e-06 ***
townTOA PAYOH            0.0256107  0.0174972   1.464 0.143406
townWOODLANDS           -0.3571252  0.0151625 -23.553  < 2e-16 ***
townYISHUN              -0.2635183  0.0148934 -17.694  < 2e-16 ***
flat_type2 ROOM          0.0864590  0.0827980   1.044 0.296490
flat_type3 ROOM          0.2562731  0.0824976   3.106 0.001916 **
flat_type4 ROOM          0.3207983  0.0846231   3.791 0.000154 ***
flat_type5 ROOM          0.3346415  0.0871193   3.841 0.000126 ***
flat_typeEXECUTIVE       0.3796448  0.0908543   4.179 3.04e-05 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1134 on 2422 degrees of freedom
Multiple R-squared:  0.8873,    Adjusted R-squared:  0.8858
F-statistic:   578 on 33 and 2422 DF,  p-value: < 2.2e-16
```

Figure 9: ANOVA Table (M_N)

```
Response: log(resale_price)
                    Df  Sum Sq Mean Sq  F value    Pr(>F)
floor_area_sqm       1 136.113 136.113 10582.09 < 2.2e-16 ***
storey_mid           1  30.817  30.817  2395.89 < 2.2e-16 ***
remaining_lease_dec  1  20.564  20.564  1598.78 < 2.2e-16 ***
town                25  56.033   2.241   174.25 < 2.2e-16 ***
flat_type            5   1.810   0.362    28.15 < 2.2e-16 ***
Residuals         2422  31.153   0.013
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

4.3 Multicollinearity Assessment

Variance Inflation Factors (VIFs) were computed to assess residual multicollinearity in the final model where Adjusted VIF (Adj. VIF) = GVIF^(1/(2×Df))

Metric	***floor_area_sqm***	***storey_mid***	***remaining_lease_dec***	***town***	***flat_type***
VIF	14.51	1.23	2.14	2.66	17.71
Adjusted VIF	3.81	1.11	1.46	1.02	1.33

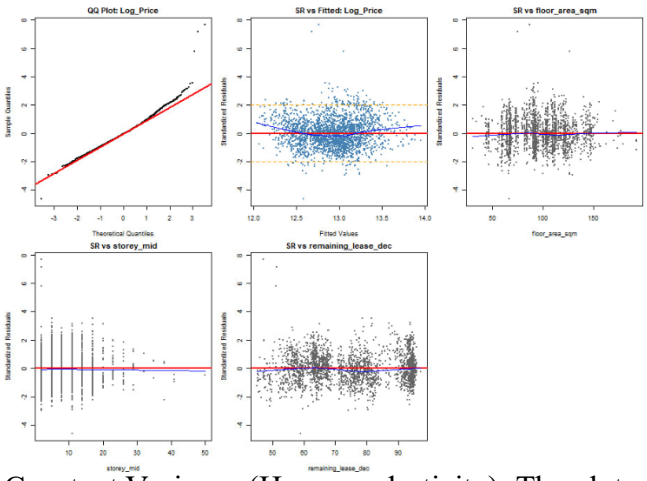
The elevated VIF for *floor_area_sqm* (14.51) reflects expected correlation with *flat_type*, as both capture aspects of flat size. However, the adjusted metric (Adj. VIF) = 3.81 remains within acceptable bounds. Both variables are retained because they provide complementary information: *flat_type* represents discrete market-tier categories with distinct buyer segments, while *floor_area_sqm* measures continuous physical dimensions. This distinction is economically meaningful, two 4-room flats of different sizes command different prices, and two flats of identical size but different types may have different values due to layout, amenities, or prestige. The remaining continuous predictors show low VIFs (< 2.2), and the town dummies exhibit minimal inflation (GVIF^(1/(2×Df)) = 1.02), confirming that multicollinearity does not undermine coefficient stability or inference. The residual standard error on the log-scale of 0.1134 indicates a small 11.34% prediction error.

4.4 Influence Diagnostics

Influence analysis confirms the model is robust to individual observations. No highly influential points were detected (maximum Cook's D = 0.089; threshold = 1). While 135 observations (5.5%) exceed the leverage threshold (h > 2p/n = 0.0277) and 12 observations (0.5%) have |Standardized Residual| > 3, verification via Cook's Distance confirms none exert undue influence on coefficient estimates. No data points require removal or special treatment.

4.5 Model Adequacy Checking

Figure 10: QQ Plot/SR vs variables plots



Residual diagnostics were conducted to verify that the final model satisfies regression assumptions.

Normality of Errors: The Q-Q plot of standardized residuals shows points closely following the reference line through the center, with o nly minor deviations in the extreme upper tail (top 1–2%). The Shapiro-Wilk test yields W = 0.9764 ($p < 0.001$). While the p-value is significant due to the large sample size (n = 2,456), the W statistic itself indicates good approximate normality. The minor tail deviations are acceptable and do not materially affect inference.

Constant Variance (Homoscedasticity): The plot of standardized residuals versus fitted values shows random scatter around zero across the entire range of fitted values. The lowess smooth line is nearly horizontal at zero, with no systematic curvature or funnel pattern. This confirms that variance is constant across low, medium, and high predicted prices, addressing the heteroscedasticity observed in the untransformed model.

Linearity: Plots of standardized residuals versus each continuous predictor (*floor_area_sqm*, *storey_mid*, *remaining_lease_dec*) show horizontal bands with flat lowess lines centered at zero. There is no evidence of curvature or systematic patterns, confirming that linear functional forms are appropriate for all continuous predictors.

Independence: The Durbin-Watson test yields D = 1.2754 ($p < 0.001$), indicating mild positive autocorrelation. This is expected in spatial data where nearby HDB towns may exhibit similar price movements. However, for cross-sectional prediction purposes, this mild autocorrelation does not materially undermine model validity or coefficient interpretation. The ANOVA F-tests confirm all predictor groups are jointly highly significant ($p < 2.2e\text{-}16$), with town (F = 174.25) and *flat_type* (F = 28.15) showing particularly strong group effects as seen in Figure 9.

Conclusion: The final model satisfies all key regression assumptions. Residual diagnostics confirm approximate normality, constant variance, linearity, and acceptable independence.

4.6 Coefficient Interpretation

For the log-transformed model, coefficients are interpreted as percentage changes in resale price via the formula: $\%\Delta price = (\exp(\beta) - 1) \times 100$.

Continuous Predictors:

Predictor	Coefficient (β)	Percentage Effect
floor_area_sqm	0.00884	**+0.888% per sqm**
storey_mid	0.00827	**+0.830% per storey**
remaining_lease_dec	0.01229	**+1.236% per year**

Town Effects (baseline: ANG MO KIO):

The town coefficients represent percentage price premiums or discounts relative to the baseline town:

Highest Premiums:	**Largest Discounts**:
1. Central Area: **+43.4%** (most expensive)	1. Sembawang: **−32.8%** (most affordable)
2. Marine Parade: **+39.7%**	2. Choa Chu Kang: **−31.0%**
3. Bukit Timah: **+37.8%**	3. Woodlands: **−20.0%**
4. Queenstown: **+25.1%**	4. Bukit Panjang: **−25.8%**
5. Bishan: **+23.9%**	5. Sengkang: **−25.5%**

These patterns reflect Singapore's well-established spatial price gradient, with central and southern towns commanding substantial premiums due to proximity to CBD, amenities, and prestige, while peripheral towns in the north and west offer more affordable housing. All three continuous predictors

are individually highly significant ($p < 2e\text{-}16$), with t-values exceeding 15, indicating precise coefficient estimates.

Flat Type Effects (baseline: 1-ROOM):

Flat Type	Percentage Premium
2-ROOM	+9.0% (not statistically significant)
3-ROOM	+29.2%
4-ROOM	+37.8%
5-ROOM	+39.7%
EXECUTIVE	+46.2%

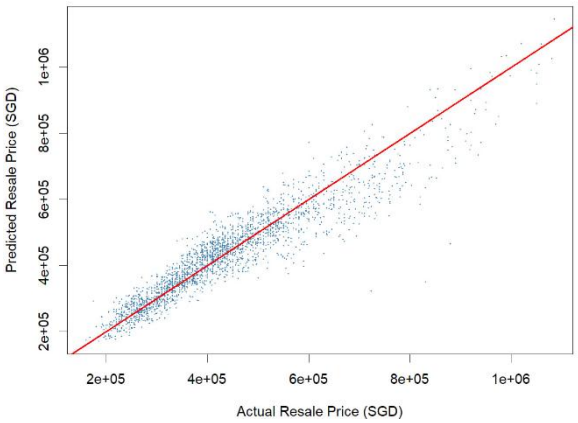
The monotonic increase from 1-room to Executive flats confirms that flat size/type is a primary price driver, with Executive flats commanding a 46% premium over 1-room units, holding location, floor area, and lease remaining constant.

4.7 Predictive Performance

In-sample predictive performance was assessed by back-transforming log-scale predictions to the original price scale and comparing against actual resale prices.

MAPE Interpretation: On average, predictions deviate from actual prices by 8.62%. This indicates good predictive accuracy for policy and planning applications, where MAPE < 10% is generally considered acceptable.

Figure 11: Predicted vs Actual



Predicted vs. Actual Plot: The scatter plot of predicted versus actual resale prices shows tight clustering around the 45° reference line across the entire price range (SGD 160,000 to SGD 1,085,000), confirming that the model performs well for both affordable and premium flats without systematic under- or over-prediction at any price level.

Part 5: Limitations and Conclusion

5.1 Limitations

The final model achieves strong fit and satisfies key regression assumptions, but several limitations should be acknowledged. First, the Durbin-Watson statistic ($D = 1.2754$) indicates mild positive autocorrelation, likely reflecting spatial clustering of nearby HDB towns with similar amenities and market dynamics. While this does not invalidate coefficient estimates, a spatial regression extension could explicitly model these dependencies. Second, all performance metrics are computed in-sample; out-of-sample validation via cross-validation would provide a more realistic assessment of predictive accuracy for deployment. Third, the elevated VIF for *floor_area_sqm* (14.51) reflects expected correlation with *flat_type*; while both are retained for complementary information, this modestly inflates the standard error for floor area. Finally, the analysis uses a single-month snapshot (July 2020); resale price drivers may shift across market cycles, and unobserved factors such as proximity to MRT stations or renovation status are not captured.

5.2 Conclusion

This analysis developed a statistically adequate log-linear regression model for predicting HDB resale prices. The final model explains 88.6% of variance in log prices (Adjusted $R^2 = 0.8858$) with good predictive accuracy (MAPE = 8.62%). Rigorous diagnostic checking confirms approximate normality, constant variance, and linearity of residuals. In summary, this analysis demonstrates a complete regression workflow: from exploratory diagnostics and transformation selection to model refinement, adequacy checking, and interpretable inference. The resulting model is both statistically rigorous and practically useful for analysis of Singapore's public housing market.